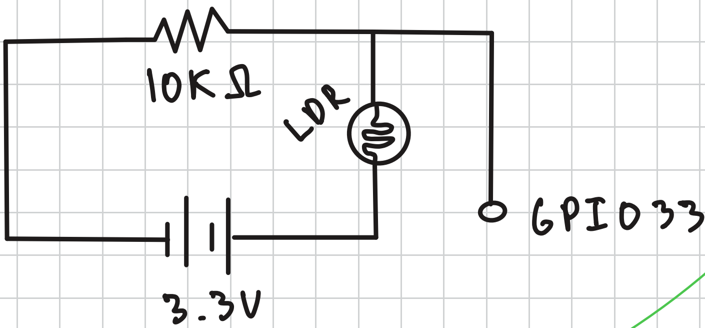
ADC → Relative Power

0. Circuit Diagram (Voltage divider)



.csv ADC data is already averaged

1. Average ADC

$$ADC_{avg} = \frac{\sum_{i=1}^{N} ADC_i}{N}$$

2. Convert ADC → Voltage

$$V_{out} = V_{in} \cdot \frac{ADC_{avg}}{4095}$$

ESP32 ADC range: 0-4095

3. Convert voltage → R_{LDR} (ADC)

In a voltage divider circuit:

$$V_{out} = V_{in}\left(\frac{R_2}{R_2+R_1}\right),$$ where $R_1 = R_{LDR}$ and

$$R_2 = 10\,k\Omega$$

Rearranging: $R_{LDR} = 10^4\,\Omega \cdot \left(\frac{V_{in}}{V_{out}} - 1\right)$

Substituting from pt. 2 $V_{out} = V_{in} \cdot \frac{ADC_{avg}}{4095}$:

$$R_{LDR} = 10^4\,\Omega \left(\frac{4095}{ADC_{avg}} - 1\right)$$

4. Find $P_{relative}$ from R_{LDR}

Rearranging the power-resistance equation of a photoresistor, $R_{LDR} = K P^{-\gamma}$, to get power:

$$P = \left(\frac{K}{R_{LDR}}\right)^{\frac{1}{\gamma}}$$

Since K is unknown for the GL5528 photoresistor, we can find a relative resistance value to get rid of K:

$$P_{ref} = \left(\frac{K}{R_{ref}}\right)^{\frac{1}{\gamma}} \Longrightarrow \frac{P}{P_{ref}} = \frac{(K/R_{LDR})^{1/\gamma}}{(K/R_{ref})^{1/\gamma}}$$

Cancelling K:

$$\frac{P}{P_{ref}} = \left(\frac{R_{ref}}{R_{LDR}}\right)^{1/\gamma} = P_{rel}$$

Note: P_{ref} is unknown above and the $\frac{P}{P_{ref}}$ ratio is equal to P_{rel}. In my Python code, I chose $R_{ref} = MIN[R_{LDR}]$, that way $P_{rel} = 1$ at the brightest point where the resistance is the lowest. Therefore, the range of P_{rel}: $(0, 1]$

5. Finalizing P_{rel} – ADC equation

From above:

$$R_{LDR} = 10^4 \left(\frac{4095}{ADC_{avg}} - 1 \right)$$

$$P_{rel} = \left(\frac{R_{ref}}{R_{LDR}} \right)^{1/\gamma}$$

$$P_{rel} = \left(\frac{R_{ref}}{10^4 \left(\frac{4095}{ADC_{avg}} - 1 \right)} \right)^{1/\gamma} \quad \ldots$$

… where $R_{ref} = MIN[R_{LDR}]$

Therefore:

$$P_{rel} = \left(\frac{MIN[R_{LDR}]}{10^4 \left(\frac{4095}{ADC_{avg}} - 1 \right)} \right)^{1/\gamma}$$